

ASHISH RAMRAKHIANI

Software Engineer | Agentic AI Engineer • Distributed Systems • Backend • Cloud-Native • Full Stack
ramrakhiyaniashish@gmail.com | +1 (541) 286-2203 | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Agentic AI and Software Engineer with 6+ years spanning enterprise, AI product, and applied engineering environments. Promoted to Technical Lead within 2 years at Colgate, directing 6 engineers on a platform sustaining 5M daily API calls; delivered 92% API latency reduction, 97% ETL improvement, and \$300K+ in annual savings. At LiliSolutions, shipped HIPAA-compliant voice AI, multi-agent RAG pipelines, and GCP observability platforms. Engineered BugSleuth, a fault-localization engine that outperformed both state-of-the-art supervised methods and LLMs without labeled training data.

PROFESSIONAL EXPERIENCE

Founding AI Software Engineer | LiliSolutions | Remote, US March 2026 – Present

- Architected a production-grade multi-agent compliance assistant (LangGraph, Python) with a self-correcting RAG loop (hybrid search via pgvector + Postgres full-text, LLM relevance grading, and query rewriting) and a Postgres checkpointer memory layer enabling durable multi-turn context and context-aware follow-up resolution; automated synthesis of 10K+ regulatory documents, reducing analyst research time by 40% under strict enterprise data governance.
- Built a HIPAA-aware patient-collection voice bot platform (Python, FastAPI, Cloud Run) integrating Retell SDK and a GPT-4o warm-transfer agent across a 25-node conversation flow, with 4 security gates (HMAC request auth, calling-window enforcement, hashed DNC opt-out, TCPA compliance), per-tenant Firestore isolation, and a 108-scenario eval harness gated in CI on critical regressions.
- Shipped a GCP-native alerting platform covering 11 production projects (including HIPAA workloads) using a Pub/Sub fan-in Cloud Function (Gen2, Python 3.12) to route SLO burn, drift, error, and cost anomaly alerts into Google Chat; eliminated per-project Terraform writes with a JSON Schema-validated YAML onboarding contract enabling single-config enrollment.

Lead Software Engineer – Cloud & AI | HipCastor Labs · Oregon State University | Corvallis, OR Sept 2024 – March 2026

- Developed an Agentic AI orchestration layer for EphFlow using LangChain to manage long-running (90+ min) multi-cloud tasks; enabled autonomous error-recovery and task decomposition reducing cloud execution costs by provisioning ephemeral VMs on-demand, validated across AWS, GCP, and GitHub Actions.
- Designed a graph-based workflow execution API with DAG validation, dependency resolution, and cross-platform job dispatch backed by language-agnostic RPC libraries (Python, R, Julia) and S3-based coordination with retry/backoff across 5 cloud backends.
- Built a React Flow + Dagre interactive DAG editor for configuring multi-cloud pipelines; implemented real-time JSON schema validation (AJV), custom node/edge rendering, and automated graph layout computation deployed via GitHub Pages with a React 19 test suite.

Research Software Engineer | ANSWER Labs · Oregon State University | Corvallis, OR Sept 2023 – Aug 2024

- Built BugSleuth, a Java 17 automated bug localization engine that combines multiple fault-signal sources using a genetic algorithm with a custom ranking metric; outperformed the prior state-of-the-art by 28% (82 more defects localized in top-5) and surpassed LLM-based localization methods across 704 real-world Java defects (Defects4J v2.0), without labeled training data.
- Engineered a configurable optimization pipeline with tunable selection, crossover, and mutation parameters; delivered millisecond-level per-defect runtime with high result stability (cv < 0.1 across 20 runs), validated on a benchmark of 704 defects spanning multiple open-source Java projects.

Software Engineer II – Technical Lead | Colgate Global Business Services | Mumbai, India Aug 2019 – Aug 2023

- Designed and owned the REST API contract end-to-end for a consumer- and retailer-facing platform serving 15 countries across 5M daily API calls; defined versioning, backward compatibility, rate limiting, and pagination as the primary integration point for 6 enterprise services. Promoted to Technical Lead within 2 years, directing 6 engineers across 3 major platform features.
- Inherited a legacy monolith with synchronous DB-per-request calls causing 12s response times; reduced latency to <1s (92%) by architecting an event-driven Redis caching tier with Kafka-based invalidation, root-caused via JVM flame graphs and APM traces.
- Cut PostgreSQL query latency by 80% by redesigning composite indexes, eliminating N+1 patterns via query plan analysis, and introducing HikariCP connection pooling; sustained 80K daily queries across 8 regional deployments.
- Redesigned a synchronous supply-chain backend into event-driven microservices using RabbitMQ Topic Exchanges with custom consumer thread pools and backpressure handling; achieved zero observed message loss across 500+ weekly product launches via at-least-once delivery and idempotent consumer design.
- Led SAP-to-Snowflake/DBT/Airflow migration using idempotent DAGs and async event-choreography; reduced processing from 8 hours to 15 minutes (97%), data-freshness from 24 hours to 15 minutes, saving \$300K+ annually across 10M records per run.

TECHNICAL SKILLS

Languages: Java 17, Python, TypeScript, JavaScript, SQL, R, Julia.

Backend Frameworks & APIs: Spring Boot, Spring Security 6, Hibernate/JPA, FastAPI, Node.js, Express.js, REST, GraphQL, gRPC, JWT/OAuth 2.0, Stripe Webhooks.

AI & Agentic Frameworks: LangGraph, LangChain, LangSmith, pgvector, RAG pipelines, MCP (Model Context Protocol), AutoGen, OpenAI Agents SDK, Vercel AI SDK, Retell SDK.

Frontend: React, Redux, Next.js 15, Tailwind CSS, shadcn/ui, React Flow, Vercel AI SDK.

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis, Snowflake, SAP HANA, Firestore; query plan analysis, composite indexing, HikariCP connection pooling, schema design.

Messaging & Data Pipelines: Apache Kafka, RabbitMQ, Airflow, DBT, ETL/ELT, idempotent DAG orchestration, backpressure handling.

Cloud & Infrastructure: AWS (Lambda, EC2, S3), GCP (Cloud Run, Cloud Functions Gen2, BigQuery, Pub/Sub), Azure, Docker, Kubernetes, Terraform, GitHub Actions, Maven, CI/CD.

Observability & Security: Prometheus, Grafana, JVM flame-graph profiling, distributed tracing, APM, AES-256.

Architecture & Reliability: Microservices, Event-Driven Architecture, Distributed Systems, System Design, API versioning, circuit breakers, retry/backoff, rolling deployments.

SELECTED PROJECTS

CareerMind: AI Job Search Platform | Next.js 15, TypeScript, Prisma, Vercel AI SDK, Docker | [GitHub](#)

- Self-hosted AI job search platform with streaming LLM resume tailoring (server-side LaTeX), provider-agnostic LLM integration (Ollama, OpenAI, DeepSeek), cron-driven job scoring pipeline, Gmail OAuth outreach with email verification (Prospero + ZeroBounce), AES-256 credential storage, Connections Kanban, and E2E-tested Docker Compose deployment.

Tome Raider: Production Agentic RAG | LangGraph, FastAPI, pgvector, Next.js 16, Supabase | [GitHub](#) | [Live Demo](#)

- Built a production-grade, multi-tenant agentic RAG system with self-correcting LangGraph retrieval loop (hybrid search + RRF + LLM reranking), history-aware query contextualization, prompt-injection screening, PII masking, per-user Postgres RLS, stateless ES256/JWKS auth, response caching, and LangSmith observability; deployed on Render via IaC Blueprint with primary/fallback model routing.

EDUCATION

Oregon State University | M.S. Computer Science | GPA: 3.96/4.0 | Corvallis, OR 2026

K.J. Somaiya College of Engineering | B.Tech. Computer Engineering | GPA: 9.26/10 | Mumbai, India 2019